

CSC633 MACHINE LEARNING



TEXAS A&M UNIVERSITY

Department of Computer
Science & Engineering

CSCE 633: Machine Learning

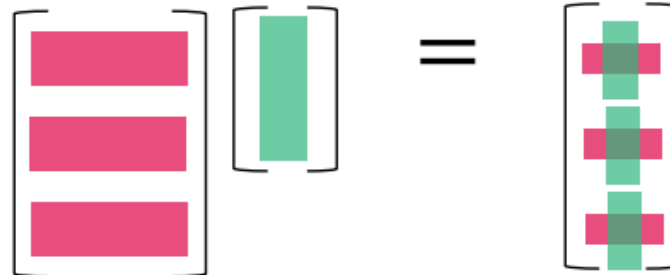
Lecture 2: Measures of Model Performance

Measure of Model Performance: Topics

- Linear algebra review
- Supervised model introduction
- Metrics of performance
- Be able to create receiver operating characteristic curves
- kNN – a simple supervised learning model

Matrix times Vector -2 Ways

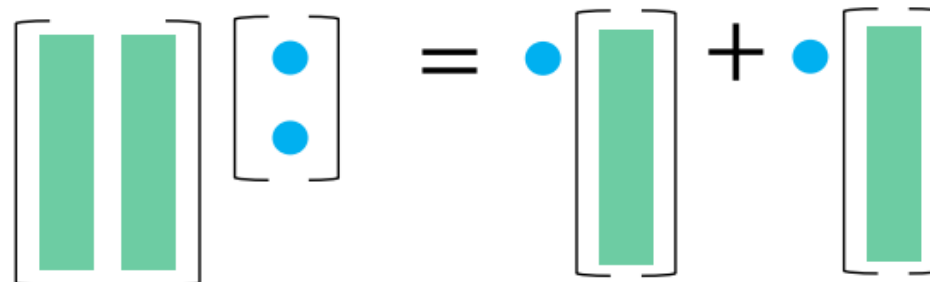
Mv1



The row vectors of A are multiplied by a vector $\mathbf{x}$ and become the three dot-product elements of $A\mathbf{x}$.

$$A\mathbf{x} = \begin{bmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} (x_1 + 2x_2) \\ (3x_1 + 4x_2) \\ (5x_1 + 6x_2) \end{bmatrix}$$

Mv2



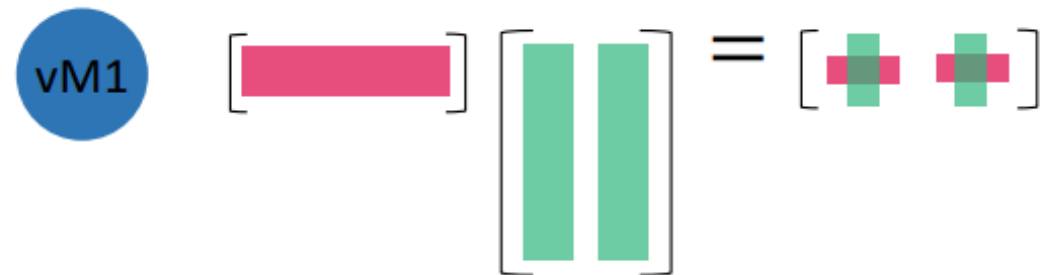
The product $A\mathbf{x}$ is a linear combination of the column vectors of A .

$$A\mathbf{x} = \begin{bmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = x_1 \begin{bmatrix} 1 \\ 3 \\ 5 \end{bmatrix} + x_2 \begin{bmatrix} 2 \\ 4 \\ 6 \end{bmatrix}$$

At first, you learn (Mv1). But when you get used to viewing it as (Mv2), you can understand $A\mathbf{x}$ as a linear combination of the columns of A . Those products fill the column space of A denoted as $C(A)$. The solution space of $A\mathbf{x}=0$ is the nullspace of A denoted as $N(A)$

Vector times Matrix – 2 Ways

vM1

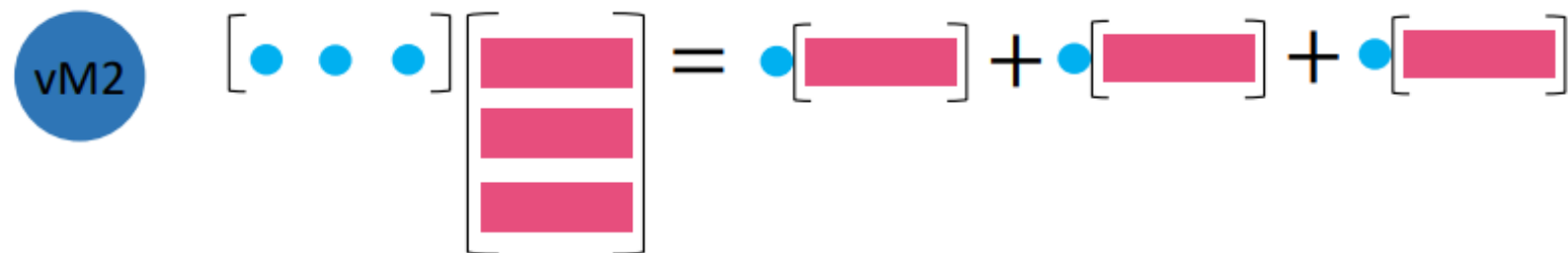


$$\begin{bmatrix} \text{pink rectangle} \end{bmatrix} \begin{bmatrix} \text{green rectangle} & \text{green rectangle} \end{bmatrix} = \begin{bmatrix} \text{pink rectangle with green cross} & \text{pink rectangle with green cross} \end{bmatrix}$$

$$\mathbf{y}A = [y_1 \quad y_2 \quad y_3] \begin{bmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{bmatrix} = [(y_1 + 3y_2 + 5y_3) \quad (2y_1 + 4y_2 + 6y_3)]$$

A row vector $\mathbf{y}$ is multiplied by the two column vectors of A and become the two dot-product elements of $\mathbf{y}A$.

vM2



$$\begin{bmatrix} \text{blue dot} & \text{blue dot} & \text{blue dot} \end{bmatrix} \begin{bmatrix} \text{pink rectangle} \\ \text{pink rectangle} \\ \text{pink rectangle} \end{bmatrix} = \text{blue dot} \begin{bmatrix} \text{pink rectangle} \end{bmatrix} + \text{blue dot} \begin{bmatrix} \text{pink rectangle} \end{bmatrix} + \text{blue dot} \begin{bmatrix} \text{pink rectangle} \end{bmatrix}$$

$$\mathbf{y}A = [y_1 \quad y_2 \quad y_3] \begin{bmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{bmatrix} = y_1[1 \quad 2] + y_2[3 \quad 4] + y_3[5 \quad 6]$$

The product $\mathbf{y}A$ is a linear combination of the row vectors of A .

Matrix times Matrix – 4 Ways

MM 1

Every element becomes a dot product of row vector and column vector.

$$\begin{bmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{bmatrix} \begin{bmatrix} x_1 & y_1 \\ x_2 & y_2 \end{bmatrix} = \begin{bmatrix} (x_1+2x_2) & (y_1+2y_2) \\ (3x_1+4x_2) & (3y_1+4y_2) \\ (5x_1+6x_2) & (5y_1+6y_2) \end{bmatrix}$$

MM 2

Ax and Ay are linear combinations of columns of A .

$$\begin{bmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{bmatrix} \begin{bmatrix} x_1 & y_1 \\ x_2 & y_2 \end{bmatrix} = A \begin{bmatrix} x & y \end{bmatrix} = \begin{bmatrix} Ax & Ay \end{bmatrix}$$

MM 3

The produced rows are linear combinations of rows.

$$\begin{bmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{bmatrix} \begin{bmatrix} x_1 & y_1 \\ x_2 & y_2 \end{bmatrix} = \begin{bmatrix} a_1^* \\ a_2^* \\ a_3^* \end{bmatrix} X = \begin{bmatrix} a_1^* X \\ a_2^* X \\ a_3^* X \end{bmatrix}$$

MM 4

Multiplication AB is broken down to a sum of rank 1 matrices.

$$\begin{bmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{bmatrix} \begin{bmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{bmatrix} = \begin{bmatrix} a_1 & a_2 \end{bmatrix} \begin{bmatrix} b_1^* \\ b_2^* \end{bmatrix} = a_1 b_1^* + a_2 b_2^*$$

$$= \begin{bmatrix} 1 \\ 3 \\ 5 \end{bmatrix} \begin{bmatrix} b_{11} & b_{12} \end{bmatrix} + \begin{bmatrix} 2 \\ 4 \\ 6 \end{bmatrix} \begin{bmatrix} b_{21} & b_{22} \end{bmatrix} = \begin{bmatrix} b_{11} & b_{12} \\ 3b_{11} & 3b_{12} \\ 5b_{11} & 5b_{12} \end{bmatrix} + \begin{bmatrix} 2b_{21} & 2b_{22} \\ 4b_{21} & 4b_{22} \\ 6b_{21} & 6b_{22} \end{bmatrix}$$

Practical Patterns (1/2)

P1'

$$\begin{bmatrix} \text{green bar} & \text{green bar} & \text{green bar} \end{bmatrix} \begin{bmatrix} \bullet & & \\ & \bullet & \\ & & \bullet \end{bmatrix} = \begin{bmatrix} \text{green bar} & \text{green bar} & \text{green bar} \end{bmatrix}$$

Applying a diagonal matrix from the right scales each column.

$$AD = [\mathbf{a}_1 \quad \mathbf{a}_2 \quad \mathbf{a}_3] \begin{bmatrix} d_1 & & \\ & d_2 & \\ & & d_3 \end{bmatrix} = [d_1 \mathbf{a}_1 \quad d_2 \mathbf{a}_2 \quad d_3 \mathbf{a}_3]$$

P2'

$$\begin{bmatrix} \bullet & & \\ & \bullet & \\ & & \bullet \end{bmatrix} \begin{bmatrix} \text{pink bar} \\ \text{pink bar} \\ \text{pink bar} \end{bmatrix} = \begin{bmatrix} \bullet & \text{pink bar} \\ \bullet & \text{pink bar} \\ \bullet & \text{pink bar} \end{bmatrix}$$

Applying a diagonal matrix from the left scales each row.

$$DB = \begin{bmatrix} d_1 & & \\ & d_2 & \\ & & d_3 \end{bmatrix} \begin{bmatrix} \mathbf{b}_1^* \\ \mathbf{b}_2^* \\ \mathbf{b}_3^* \end{bmatrix} = \begin{bmatrix} d_1 \mathbf{b}_1^* \\ d_2 \mathbf{b}_2^* \\ d_3 \mathbf{b}_3^* \end{bmatrix}$$

Practical Patterns (2/2)

P3

$$\begin{bmatrix} \text{green bar} & \text{green bar} & \text{green bar} \end{bmatrix} \begin{bmatrix} \bullet & & \\ & \bullet & \\ & & \bullet \end{bmatrix} \begin{bmatrix} \bullet \\ \bullet \\ \bullet \end{bmatrix} = \begin{bmatrix} \bullet & \text{green bar} \end{bmatrix} + \begin{bmatrix} \bullet & \text{green bar} \end{bmatrix} + \begin{bmatrix} \bullet & \text{green bar} \end{bmatrix}$$

This pattern makes another combination of columns.
You will encounter this in differential/recurrence equations.

$$XD\mathbf{c} = \begin{bmatrix} x_1 & x_2 & x_3 \end{bmatrix} \begin{bmatrix} d_1 & & \\ & d_2 & \\ & & d_3 \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \\ c_3 \end{bmatrix} = c_1 d_1 x_1 + c_2 d_2 x_2 + c_3 d_3 x_3$$

P4

$$\begin{bmatrix} \text{green bar} & \text{green bar} & \text{green bar} \end{bmatrix} \begin{bmatrix} \bullet & & \\ & \bullet & \\ & & \bullet \end{bmatrix} \begin{bmatrix} \text{pink bar} \\ \text{pink bar} \\ \text{pink bar} \end{bmatrix} = \begin{bmatrix} \bullet & \text{pink bar} \\ \text{green bar} & \text{gray} \end{bmatrix} + \begin{bmatrix} \bullet & \text{pink bar} \\ \text{gray} & \text{green bar} \end{bmatrix} + \begin{bmatrix} \bullet & \text{pink bar} \\ \text{gray} & \text{pink bar} \end{bmatrix}$$

A matrix is broken down to a sum of rank 1 matrices,
as in singular value/eigenvalue decomposition.

$$U\Sigma V^T = \begin{bmatrix} \mathbf{u}_1 & \mathbf{u}_2 & \mathbf{u}_3 \end{bmatrix} \begin{bmatrix} \sigma_1 & & \\ & \sigma_2 & \\ & & \sigma_3 \end{bmatrix} \begin{bmatrix} \mathbf{v}_1^T \\ \mathbf{v}_2^T \\ \mathbf{v}_3^T \end{bmatrix} = \sigma_1 \mathbf{u}_1 \mathbf{v}_1^T + \sigma_2 \mathbf{u}_2 \mathbf{v}_2^T + \sigma_3 \mathbf{u}_3 \mathbf{v}_3^T$$

Measure of Model Performance: Topics (end)

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